"Earth Processes"

Plate Tectonics

Introduction:

Earthquakes occur because of the force i.e. stress happening in rocks. These dynamic internal forces collectively known as diastrophism, tend to cause the earth's surface to undergo deformation. Tectonic movement creates tremendous stress near plate boundaries; this stress deforms rocks. (Olivar, 2016) (Turk & Thompson, 2012)

I. FACTORS THAT CONTROL HOW A ROCK RESPONDS TO STRESS

A. Nature of a Rock

Think of a quartz crystal, a gold nugget, and a rubber ball. If you strike quartz with a hammer, it shatters- that is, it fails by brittle fracture. In contrast, if you strike the gold nugget, it deforms in a plastic manner- it flattens and stays flat. If you hit the rubber ball, it deforms elastically and rebounds immediately, sending the hammer flying back at you. Initially, all rocks react to stress by deforming elastically by a slight amount. Near Earth's surface, where temperature and pressure are low, different types of rocks behave differently with continuing stress. Granite and quartzite tend to fracture in a brittle manner. Other rocks, such as shale, limestone, and marble, tend to deform plastically.

B. Temperature

The higher the temperature, the greater the tendency of a rock to deform in a plastic manner. It is difficult to bend an iron bar at room temperature, but if a bar is heated to a red-hot temperature, it becomes plastic and bends easily.

C. Pressure

High non-directed (hydrostatic) pressure favors plastic behavior. Thus, because both temperature and pressure increase during burial, and because both of these factors promote plastic deformation, deeply buried rocks have a greater tendency to bend and flow under stress than do shallow rocks.

D. Time

Stress applied slowly, rather than suddenly, also favors plastic behavior. Marble park benches in New York City have sagged plastically underweight within 100 years. In contrast, rapidly applied stress, such as the blow of a hammer, causes brittle fracture in a marble bench. (Turk & Thompson, 2012)

In other words, for deformation to occur, the following conditions must be met: (Olivar, 2016)

1. The rock material must have the ability to deform under pressure and heat.
2. The higher the temperature of the rock the more elastic it becomes.
3. Pressure must not exceed the internal strength of the rock otherwise, fracturing occurs.
4. Deformation must be applied slowly.

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II. GEOLOGIC STRUCTURES

A geologic structure is any feature produced by rock deformation. A rock has been deformed has been:

- translated (moved) from its original position
- changed in orientation
- changed in shape

There are three types of geologic structures; folds, faults, and joints. When deformation is ductile, rocks bend without breaking, forming folds. When deformation is brittle, rocks break, forming joints or faults.

A. Folds

A fold is a bend in rock. Some folded rocks display **little or no fracturing**, indicating that the rocks deformed in a plastic manner. In other cases, folding occurs by a combination of **plastic deformation and brittle fracture**. Folds formed in this manner exhibit many tiny fractures.

i. Characteristics of Folds

If you hold a sheet of clay between your hands and squeeze your hands together, the clay deforms into a sequence of folds. This demonstration illustrates three characteristics of folds:

a. Folding usually results from compression. For example, tightly folded rocks near Palmdale, California, indicate that the region was compressed as tectonic plates converged.
b. Folding always shortens the horizontal distances in rock.
c. A fold usually occurs as part of a group of many similar folds.

ii. Common Types of Folds

A fold arching upward is an anticline and one arching downward is a syncline. The sides of a fold are called the limbs.

Anticlines are most commonly caused by compressive stress. Geologists recognize anticlines in the field by 1) seeing strata that dip away from the axis of the fold; 2) noting that erosion eposes progressively older sediments as you move towards the fold axis. Even though an anticline is structurally a high point in a fold, anticlines do not necessarily form **topographic** ridges. Conversely, synclines **do not** form valleys. Landforms are created by combinations of tectonic and surface processes. A syncline lies beneath the peak and an anticline forms a saddle between two peaks simply because some portions of the folded rock eroded faster than others. *Synclines are like basins* – they are U-shaped folds in which strata dip towards the fold axis and the strata become younger as one moves towards the fold axis. A circular or elliptical anticlinal structure is called a dome. A dome resembles an inverted cereal bowl. Sedimentary layering dips away from the top of a dome in all directions. A similarly shaped syncline is called a basin. Domes and basins can be small structures only a few kilometers in diameter or less. Alternatively, a large basin or dome can be 500 kilometers in diameter. These huge structures form by the sinking or rising of continental crust in response to vertical movements of the underlying mantle.

Cylindrical anticlines (perfectly shaped) are rare in nature. Most anticlines (and synclines) "plunge" into the Earth at an angle.

A monocline is an open, step-like structure in which the layers are all inclined in the same direction on either side of the fold axis. On the Colorado Plateau, there are broad areas of nearly flat lying strata (plateaus) separated by bends along monoclinal folds.

B. Faults

A fault is a fracture along which rock on one side moved relative to rock on the other side. Slip is the distance that rocks on opposite sides of fault have moved. Movement along a fault may be gradual, or the rock may move suddenly, generating an earthquake. Some fault occur as single fractures in rock; most, however, consist of numerous closely spaced fractures within a fault zone. Rock may slide hundreds of meters or many kilometers along a large fault zone.

Why does a rock move repeatedly along many faults and fault zones?

1. Tectonic forces commonly persist in the same place over long periods of time
2. Once a fault forms, it is easier for rock to move again along the same fracture than for a new fracture to develop nearby.

A **normal fault** forms where tectonic movement stretches Earth's crust, pulling it apart. As the crust stretches and fractures, the hanging wall moves down relative to the footwall.

In a region where compressive forces squeeze the crust, geologic structures must accommodate crustal shortening. A fold accomplishes shortening. Normal faults most often form due to tensional stress. Uplifting a broad region like the Colorado Plateau stretched the brittle crust, creating tension which fractured the strata. A number of north-south trending normal faults dissect the Colorado Plateau. At Parashant, the huge, nearly vertical fault scarps that make up the Hurricane Cliffs and Grand Wash Cliffs are dramatic examples of this structural feature.

Alternatively, compressive forces may fracture the rock to produce a **reverse fault**. Here, the hanging wall has moved upward relative to the footwall, as tectonic force squeezed the rock horizontally. Reverse faults form most often due to compression – forces that push rock together, breaking the rock and creating a fault. Low-angle (< 30 degrees) reverse faults are called thrust faults.

What are footwall and hanging wall?

Hydrothermal solutions often precipitate rich ore veins in the fractures of a fault zone. Miners then dig shafts and tunnels along veins to get the ore. Many faults are not vertical but dip into Earth at an angle. Therefore, many veins have an upper and a lower side. Miners refer to the side that hangs **over their heads** as the hanging wall and the side **they walk on** is the footwall. These names are commonly used to describe both ore veins and faults.

A **thrust fault** is a special type of reverse fault that is nearly horizontal. A strike-slip fault is one in which the fracture is

vertical, or nearly so, and rocks on opposite sides of the fracture move horizontally past each other. A transform plate boundary is a strike-slip fault. Strike-slip faults arise due to shear stress – forces moving past one another. Strike-slip faults are typically nearly vertical and move rock horizontally past each other on either side of the fault. The San Andreas Fault in California is a famous example. When the side across the fault moves to the right, it is called a right-lateral strike-slip fault. If the side across the fault moves to the left, it is a left-lateral strike-slip fault. The famous San Andreas Fault is a zone of strike-slip faults that form the boundary between the Pacific plate and the North American plate.

C. Joints

A joint is a fracture in which rocks on either side of the fracture have not moved. Tectonic forces also creates joints. Most rocks near Earth's surface are jointed, but joints become less abundant with depth because rocks become more plastic and less prone to fracturing at deeper levels in the crust.

Joints and faults are important in **engineering, mining, and quarrying** because they are planes of weakness in otherwise strong rock. Dams constructed in jointed rock often leak, not because the dams themselves have holes but because water seeps into the joints and flows around the dam through the factures.

D. Folds, Faults, and Plate Boundaries

Each of the three types of plate boundaries- divergent, transform, or convergent – produces different tectonic stresses and therefore different kinds of folds and faults. Tectonic plates drift apart at a divergent boundary (the Mid-Oceanic Ridge systems and continental rifts), stretching adjacent rock and producing normal faults and grabens but little folding of rocks.
At a transform boundary, friction often holds rock together as the plates gradually slip past each other. The resultant stress may fold, fault, and uplift nearby rocks. Forces of this type have formed the San Gabriel Mountains along the San Andreas Fault zone, as well as mountain ranges north of the Himalayas.
Near a convergent plate boundary, compression commonly produces large regions of folds, reverse faults, and thrust faults.

Although plate convergence commonly creates horizontal compression, in some instances crustal extension and normal faulting also occurs at a convergent plate boundary.

III. THE SEA-FLOOR SPREADING HYPOTHESIS

Shortly after World War II, scientists began to explore the floors of Earth's oceans. Defense strategies wanted a detailed knowledge of sea floor topography for submarine warfare, and the same information was needed to lay undersea telephone cables.

What did they Discover when they mapped the Seafloor?

As they mapped the sea floor, oceanographers discovered the largest mountain chain on Earth, now called the Mid-Oceanic Ridge system. One leg of this huge submarine mountain chain, called the Mid-Oceanic Ridge, lies directly in the middle of the **Atlantic Ocean**, halfway between **North and South America** to the west, and **Europe and Africa to the East**.

Which are the oldest rocks?

The rock at the far left and far right (rock that has spread the farthest) of the Mid-Atlantic Ridge represent the oldest rock. This is the where magma is oozing out and forming new seafloor.

Which are the youngest rocks?

The youngest rocks are in the middle of Mid-Atlantic Ridge. This is the area of the ocean floor that formed most recently from magma oozing out of the Mid-Atlantic Ridge.

Where are the youngest rocks?

The youngest rocks are immediately adjacent to the Mid-Atlantic Ridge.

You learned that oceanic crust is composed mostly of basalt, an igneous rock rich in iron. As basaltic lava cools and becomes solid rock, the iron-rich minerals become weak magnets that align their magnetic fields parallel to Earth's magnetic field. Thus, basalt records the orientation of Earth's magnetic field at the time the rock cools.

In addition to mapping the topography of the sea floor, oceanographers towed devices called *magnetometers* behind their research vessels to detect and record magnetic patterns in the deep oceans. The sea floor is found to have **alternating normal and reversed polarity**, a pattern that is **symmetrically** distributed about the Mid-Oceanic Ridge.

In the mid-1960s, three scientists- *Frederick Vine, Drummond Matthews,* and *Lawrence Morley*- proposed an explanation for these odd magnetic patterns on the sea floor. They knew that scientist had recently discovered that Earth's magnetic field has reversed its polarity on the average of every 500,000 years. When such a magnetic reversal of Earth's field occurs, the **north magnetic pole becomes the south magnetic pole**, and vice versa. This discovery surprised most geologists, but periodic reversals of the Earth's magnetic field were well known by the early 1960s.

Vine, Matthews, and Morley suggested that the sea floor is spreading continuously away from the Mid-Oceanic Ridges, like two **conveyor belts** moving outward, away from each other. New basalt lava rises cracks that form at the ridge axis as the two sides of the sea floor separate. As the lava cools and solidifies, the basalt acquires the magnetic orientation of Earth's field. Because the Earth's field **periodically reverses**, the sea floor acquires a striped magnetic orientation. Thus, the sea floor and oceanic crust should become **older** with increasing distance from the ridge axis.

Furthermore, the oceanographers found that fossils in the **deepest layers** of mud overlying basalt were **very young at the ridge axis** but become progressively older with increasing distance from the ridge. This discovery, too, indicated that the sea floor becomes older with increasing distance from the ridge axis.

Symmetrical magnetic patterns and similar mud age and thickness trends were quickly discovered at other parts of the Mid-Oceanic Ridge system in other **ocean basins**, and the hypothesis of sea-floor spreading was proposed as a general model for the origin of all oceanic crust. In a very few years, the sea floor spreading hypothesis became the **basis** for development of the much broader **theory of plate tectonics**.

IV. THE DEVELOPMENT OF PLATE TECTONICS THEORY

After geologists had developed the plate tectonics theory, they began to ask: "Why do the great **slabs** of lithosphere glide over Earth's surface? Research shows that **subduction** continues slowly all the way to the **core-mantle boundary**, to a depth of 2,900 kilometers. At the same time, equal volumes of hot rock **rise** from the deep mantle to the surface beneath a spreading center (divergent boundary), forming new lithosphere to replace that lost to subduction. Recall that convection refers to the upward and downward flow of fluid material in response to heating and cooling. The process of mantle convection **continually stirs** the entire mantle as rock that is hotter than its surroundings rises toward Earth's surface and old plates that are colder than their surroundings sink into the mantle. A single mantle convection cell may be thousands of kilometers across. In this way, the entire mantle- lithosphere system circulates in great cells, carrying rock from the core- mantle boundary to Earth's surface and then back into the deepest mantle.

Rising temperature causes most materials, including soup (or rock), to expand. When soup at the bottom of the pot is heated by the underlying stove, it becomes warm and expands. It then rises because it is less dense than the soup at the top. When the hot soup reaches the top of the pot, it flows along the surface until it cools and sinks.

Just as the soup on the stove rests on the hot burner, the base of the mantle lies on- and is heated by – the hotter core. Thus, heat from the core, supplemented by additional heat generated by radioactivity within the mantle, drives the entire mantle and the lithosphere in huge cells of convecting rock. A tectonic plate is the upper portion of a convecting cell and thus glides over the asthenosphere as a result of the convection.

Two other processes, may facilitate the movement of tectonic plates. Notice that the base of the **lithosphere slopes downward** from a spreading center; the grade can be as steep as eight (8) percent, steeper than most paved highways. Calculations show that even if the slope were less steep, **gravity** would cause the lithosphere to slide away from a spreading center over the soft, plastic asthenosphere at a rate of a few centimeters per year. In addition, the **lithosphere becomes denser than the asthenosphere** below. Consequently, it can no longer float on the asthenosphere and sinks into the **mantle in a subduction zone,** dragging the trailing portion of the plate over the asthenosphere. Both of these processes may contribute to the movement of a lithospheric plate as it glides over the asthenosphere.

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